

High Altitude Long Endurance (HALE) UAV Sustained Flight through Renewable Energy

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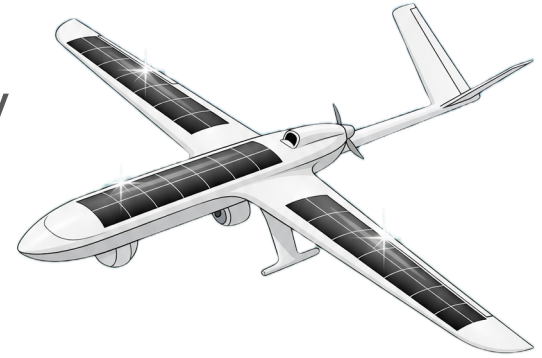
Project Overview

Objective: Theoretical design and feasibility evaluation of a solar-powered HALE UAV.

Focus Areas: Aerodynamics, structure, energy systems, and mission performance.

Motivation: Need for long-endurance, high-altitude platforms for surveillance, communication, and atmospheric monitoring.

Approach: Literature-backed modelling and system-level design specification.



Market Survey

Relevance of Research in
Current Markets

Applications:

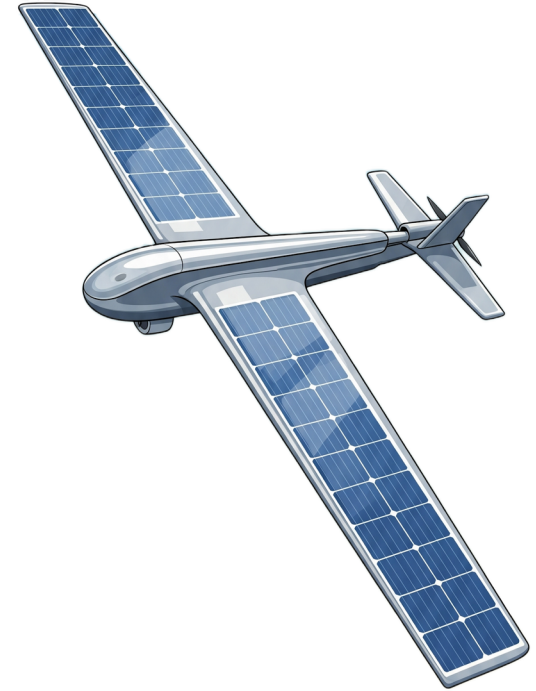
- Persistent ISR, border surveillance
- Climate and atmospheric monitoring
- Communication relays (HAPS)
- Disaster observation

Observations:

- Global push toward satellite alternatives
- Growing interest in solar/electric HALE platforms
- Demand for multi-day endurance with minimum operational cost
- Leading programs: Zephyr, Global Hawk, Helios

Literature Review

1. Aerodynamic challenges at low Reynolds numbers
2. High-aspect ratio wings essential for efficiency
3. Structural reliance on CFRP composites for stiffness & low weight
4. Solar cells: triple-junction (25–30% efficiency)
5. Energy storage bottleneck: night-time endurance
6. Fuel cells superior to batteries when energy demand >2.8 kWh
7. Joined-wing configuration highly efficient (span efficiency 1.4–1.5)



Problem Definition

Current HALE systems face:

Insufficient night-time power storage

Aerodynamic penalties at low Re regimes

Flutter and structural deformation in ultra-light wings

Limited hybrid solar–battery–fuel cell integration

Environmental sensitivity
(temperature, irradiance, density)

Design and theoretically validate a solar-powered HALE UAV capable of sustained high-altitude flight by optimizing aerodynamic configuration, structural layout, and renewable energy integration.

The theoretical model

Conceptual HALE UAV design is feasible in terms of aerodynamics, solar power generation, and night-time energy storage.

Primary Outputs:

1. Wing area: 32.0 m^2 ; solar-covered area (95%): 30.4 m^2 .
2. Aircraft operational mass excluding energy storage: 114.4 kg (weight = $1,121.9 \text{ N}$).
3. Cruise propulsive power required: $1,346 \text{ W}$.
4. Electrical power required at system level (including propulsive and electronics losses): $1,994 \text{ W}$.
5. Incident irradiance at 18–20 km (assumed transmittance 0.90): $1,224.9 \text{ W/m}^2$.
6. Solar electrical generation (28% cell eff., after electronics): $\approx 9,384 \text{ W}$.
7. Night energy required for 12 hours: $\approx 23,933 \text{ Wh}$ (23.93 kWh).
8. Required Li-ion battery mass (250 Wh/kg): $\approx 95.7 \text{ kg}$.
9. Required hydrogen (LHV 33.3 kWh/kg , fuel cell system eff 50%): $\approx 1.44 \text{ kg H}_2$. With a conservative hydrogen system mass multiplier ($\times 3$ to account for tanks and BOP), hydrogen system total mass $\approx 4.31 \text{ kg}$.

Sensitivity:

Solar generation scales with wing area and solar cell efficiency; even at 20% cell efficiency the daytime solar power greatly exceeds cruise power.

Battery mass required for 12-hour night remains large ($\sim 96 \text{ kg}$) under the current assumptions and would be impractical as a primary stored-energy solution for a 114 kg airframe.

Using hydrogen fuel cells reduces stored-energy mass dramatically: $\sim 1.4 \text{ kg H}_2$ required for 23.9 kWh ($\approx 1.44 \text{ kg H}_2$), and with system mass multiplier $\approx 4.3 \text{ kg}$ total—far lighter than batteries.

Design Specifications

Mission Requirements:

Altitude: $\geq 18\text{--}20$ km

Endurance: ≥ 24 hours (target multi-day)

Cruise at low speed for max L/D

Lightweight

scientific/communication

payload

System Requirements:

High-aspect ratio wing
(biplane / BWB / joined-wing)

CFRP composite structure

Triple-junction solar array + Li-ion or
fuel cell storage

Stable longitudinal & directional control
at low air density

Energy System & Performance Modelling

Solar Modelling:

Triple-junction cells (25–30% efficiency)

Solar irradiance estimation at 18–20 km

Wing-area-to-power ratio studies

Energy Storage:

Battery discharge modelling

Fuel cell feasibility (superior for >2.8 kWh)

Day–night energy balance via MATLAB/Simulink

Performance:

Power requirement increases by 3.5% per km altitude

Energy sustainability validated through theoretical cycle simulation

Key Findings

- Hybrid solar and battery/fuel cell system theoretically supports continuous operation
- Joined-wing configuration offers best aerodynamic-structural trade-off
- Structural analysis supports CFRP-based ultra-light design
- Theoretical modelling indicates feasible daytime power surplus and manageable night-time deficit with optimized storage
- Performance trends align strongly with validated HALE platforms in literature

Conclusion

The project establishes the feasibility of a solar-powered HALE UAV concept by integrating aerodynamic, structural, and energy-system insights. Theoretical evaluation supports long-endurance capability using optimized wing geometry and hybrid energy systems.

Future Work

High-fidelity CFD and FEM simulations
Scaled prototype fabrication
Real-time atmospheric modelling
Experimental testing of wing sections & hybrid energy systems
Development of onboard energy management algorithms